

Internet of Things Laboratory (EC2601-1)

A Project report on

**IoT-Based Fire and Smoke Detection System Using Arduino
and GSM Notification Module**

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CERTIFICATE

*This is to certify that **Sumanth (NNM23AC022), Ganesh Nagure (NNM23AC023), H S Adithyashyama (NNM23AC024) , Hamshika S Shetty (NNM23AC025)** , bonafide students of N.M.A.M. Institute of Technology, Nitte have submitted the report for the project entitled **“IOT-BASED FIRE AND SMOKE DETECTION SYSTEM USING ARDUINO AND GSM NOTIFICATION MODULE”** in partial fulfillment of the requirements for the Internet of Things Laboratory (EC2601-1) during the year 2025-2026.*

Project Evaluation

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ABSTRACT

This project presents an IoT-based Fire and Smoke Detection System designed to provide early warning during hazardous situations. The system uses an MQ-2 smoke sensor and a flame sensor to continuously monitor the environment for gas leakage, smoke, or fire. When abnormal levels are detected, the Arduino controller immediately activates a buzzer and sends real-time SMS alerts to the user through a GSM module. This ensures quick response, even when the user is away from the location. The system is low-cost, reliable, and suitable for homes, laboratories, industries, and commercial buildings, offering an efficient method of enhancing safety through IoT-enabled monitoring.

This project focuses on the design and development of an IoT-based Fire and Smoke Detection System using Arduino, aimed at providing rapid and reliable alerts during hazardous fire incidents. The system integrates an MQ-2 sensor for detecting smoke and combustible gases, along with a flame sensor for identifying fire intensity through infrared radiation. These sensors continuously monitor the surrounding environment and send real-time data to the Arduino microcontroller for analysis. When smoke concentration or flame presence crosses a predefined threshold, the system immediately triggers a local alarm using a buzzer and simultaneously sends an emergency SMS notification to the registered mobile number through the GSM module (SIM800L/SIM900). This IoT-enabled alert mechanism ensures that users are informed instantly, even when they are not physically present at the location of the incident. The proposed system offers a cost-effective, scalable, and energy-efficient solution suitable for homes, industries, laboratories, and commercial buildings. By enabling early detection and rapid response, the system significantly reduces the risk of property damage and enhances overall fire safety management.

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ABBREVIATIONS

Abbreviation	Full Form
IoT	Internet of Things
GSM	Global System for Mobile Communication
SMS	Short Message Service
LCD	Liquid Crystal Display
MQ-2	Methane/Butane/Smoke Gas Sensor
IR	Infrared
LED	Light Emitting Diode
VCC	Voltage Common Collector (Power Supply +)
GND	Ground (Power Supply –)
Rx	Receive Pin
Tx	Transmit Pin
ADC	Analog to Digital Converter
UNO	Arduino UNO Microcontroller Board
SIM	Subscriber Identity Module
LPG	Liquefied Petroleum Gas
PCB	Printed Circuit Board
DC	Direct Current
UART	Universal Asynchronous Receiver/Transmitter
Hz	Hertz
°C	Degree Celsius

Tabel 1 : Abbreviations

CHAPTER 1**INTRODUCTION**

Fire outbreaks and gas leakages continue to be major hazards in residential, commercial, and industrial environments. These incidents often result in severe losses due to delayed detection and slow emergency response. Traditional fire alarm systems, although effective to some extent, are mainly limited to audible alarms within the premises and fail to notify users when they are away. With the rapid growth of embedded systems and the Internet of Things (IoT), smarter and more responsive safety systems are now possible, offering improved detection accuracy and remote alert capabilities.

This project focuses on the development of an **IoT-based Fire and Smoke Detection System** using an Arduino microcontroller, MQ-2 smoke/gas sensor, flame sensor, and a GSM communication module. The system continuously monitors environmental parameters, detecting early signs of smoke, flammable gases, or the presence of fire. When sensor readings exceed the predefined safety threshold, the Arduino triggers two simultaneous actions: it activates a buzzer to alert individuals nearby, and it sends an **instant SMS notification** to a registered mobile user via the GSM module. This combination of local and remote alerting makes the system significantly more effective compared to conventional alarm systems.

The integration of IoT technology enhances the overall performance of the system by enabling **real-time monitoring, automated communication, and remote accessibility**. Using low-cost components, the proposed design is not only economical but also easy to install, making it highly suitable for households, offices, laboratories, warehouses, and factories. Additionally, the modular structure of the system allows future expansion, such as cloud data logging, integration with mobile applications, and automated fire-safety mechanisms.

The key highlights of the project include:

- **Continuous monitoring** of smoke, gas leakage, and flame levels.
- **Instant SMS alerts** through the GSM module for remote notification.
- **Arduino-based processing** for accurate, real-time decision-making.
- **Low-cost design** suitable for mass deployment.
- **Improved safety** through early detection and timely response.

Overall, this development project demonstrates how IoT and embedded systems can be effectively utilized to build smart safety solutions that enhance protection against fire-related hazards. It contributes toward the of modern, automated fire-safety infrastructure that ensures quicker action, reduces human risk, and minimizes property damage.

1.1 OBJECTIVES:

The objective of this project is to design a low-cost and reliable fire-safety monitoring system that uses MQ-2 and flame sensors to detect smoke, gas leakage, and fire in real time. Arduino serves as the main controller for processing sensor data and activating safety responses such as a buzzer alarm during emergency conditions. The system also sends instant SMS alerts to the user through a GSM module, enabling quick human action and enhancing overall safety. Additionally, the design focuses on easy installation, operation, maintenance, and provides scalability for future IoT-based expansion.

CHAPTER 2**SYSTEM ARCHITECTURE AND WORKFLOW**

The methodology describes the systematic approach used to design, develop, and implement the IoT-based Fire and Smoke Detection System. The system integrates multiple sensors, a microcontroller, and a GSM communication module to monitor environmental conditions and provide real-time alerts. The process includes hardware design, software programming, system integration, and testing.

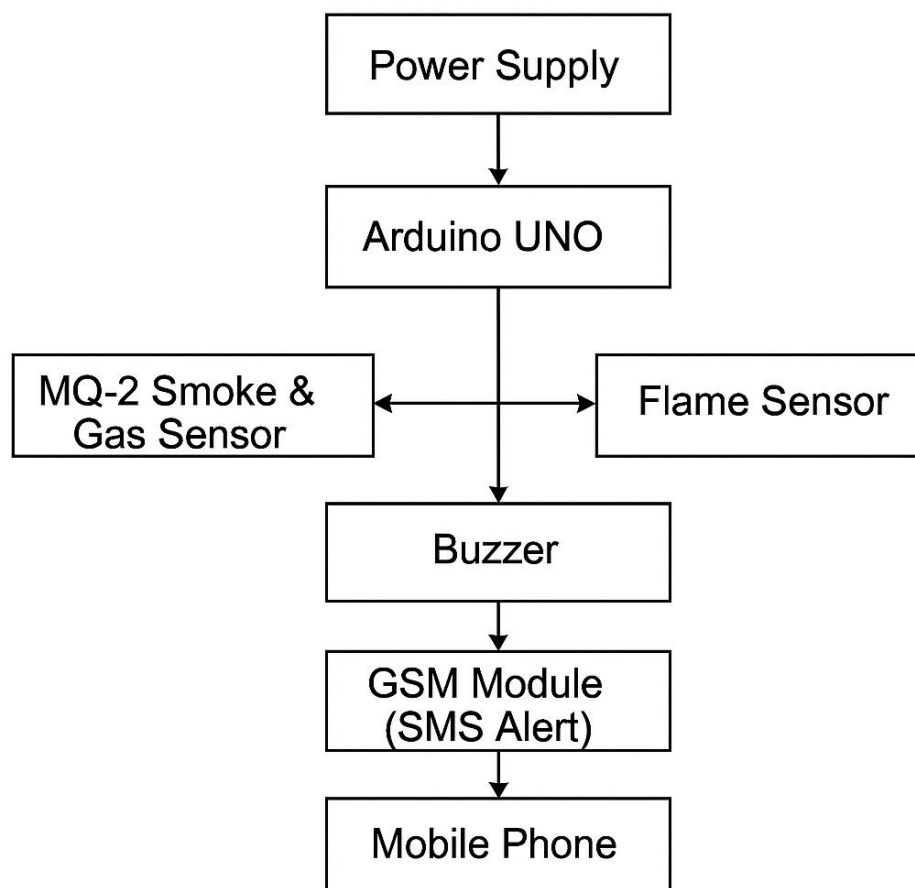
2.1 Block Diagram of the System

Figure 2.1: Block diagram of the system

The system consists of a **PSU**, an **MCU (Arduino UNO)**, **MQ-2 smoke sensor**, **IR flame sensor**, **buzzer**, and a **GSM module**, all working together to detect fire or smoke. When danger is sensed, the buzzer alerts locally while the GSM module sends an **SMS** notification to the user's mobile phone.

2.2 FSystem Operation

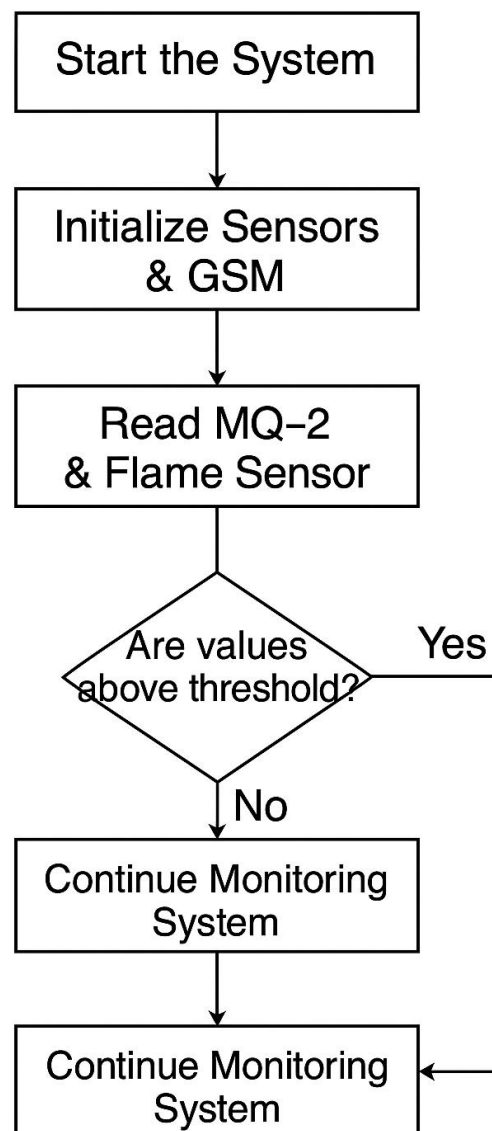


Figure 2.2: Flowchart of System Operation

The system initializes all hardware modules and continuously reads **smoke, gas, and flame sensor** values in real time. If any value crosses the threshold, the **buzzer** activates and an **SMS** alert is sent, otherwise the system keeps monitoring continuously.

2.3 Circuit Diagram

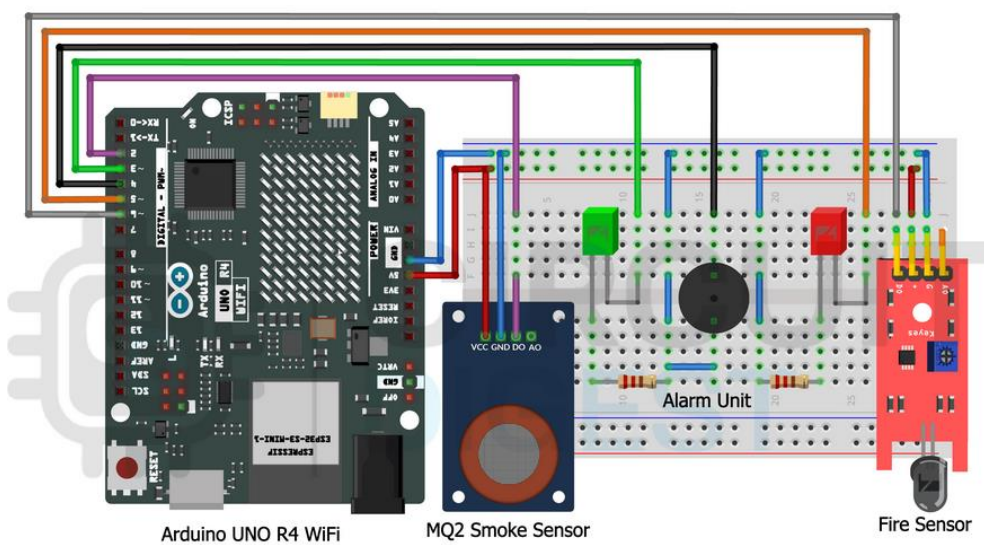


Figure 2.3: Circuit diagram

The circuit connects the Arduino **UNO R4 WiFi** with an **MQ-2** smoke sensor, flame sensor, **LEDs**, and a **buzzer** to form a complete alarm system. When smoke or fire is detected, the Arduino activates the buzzer and LEDs to provide an immediate visual and sound alert.

CHAPTER 3

COMPONENT REQUIREMENTS IOT BASED FIRE AND SMOKE DETECTION SYSTEM

The proposed fire and smoke detection system requires a combination of essential hardware and software components to ensure accurate sensing, reliable processing, and effective alert communication. The hardware components include sensors for detecting smoke and flames, a microcontroller for processing data, and output devices such as a buzzer and LEDs for signaling emergency conditions. Complementing the hardware, the software tools provide the necessary environment for programming, debugging, and establishing communication with external modules like GSM or WiFi. Together, these hardware and software requirements enable the system to function efficiently, ensuring real-time monitoring and timely alerts during hazardous situations.

3.1: HARDWARE REQUIREMENTS

Sl. No.	Hardware Component	Description
1	Arduino UNO / Arduino Nano	Main microcontroller used to read sensor data and control the alert system.
2	MQ-2 Smoke/Gas Sensor	Detects smoke and combustible gases for early warning.
3	Flame Sensor (IR Detector)	Identifies the presence of flame within its IR detection range.
4	ESP8266 / GSM Module (SIM800L/SIM900A)	Sends SMS alerts or notifications to the user.
5	Buzzer	Provides audible alarm during emergency conditions.
6	LED Indicators	Visual indication of system status or warnings.
7	5V Power Supply / USB Cable	Supplies regulated power to the microcontroller and sensors.
8	Jumper Wires	Used for electrical connections between components.
9	Breadboard / PCB	For assembling and mounting the circuit.
10	Resistors (as required)	Used for current limiting and protection of LEDs/sensors.

Tabel 2 : Hardware requirements

3.2: SOFTWARE REQUIREMENTS

SL. No.	SOFTWARE / TOOL	DESCRIPTION
1	ARDUINO IDE	USED FOR WRITING, COMPILING, AND UPLOADING CODE TO THE ARDUINO MICROCONTROLLER.
2	ESP8266 / GSM LIBRARY	ENABLES COMMUNICATION BETWEEN ARDUINO AND WIFI/GSM MODULES FOR SENDING SMS ALERTS.
3	SENSOR LIBRARIES (OPTIONAL)	HELPS IN EFFICIENT READING AND PROCESSING OF MQ-2 AND FLAME SENSOR DATA.
4	ARDUINO USB DRIVERS	REQUIRED FOR THE COMPUTER TO DETECT AND COMMUNICATE WITH THE ARDUINO BOARD.
5	SERIAL MONITOR (ARDUINO IDE)	USED FOR VIEWING SENSOR READINGS, DEBUGGING ISSUES, AND CALIBRATING THRESHOLDS.
6	FIRMWARE FOR WIFI/GSM MODULE	SUPPORTS AT COMMANDS FOR SMS TRANSMISSION AND NETWORK CONNECTIVITY.
7	OPERATING SYSTEM (WINDOWS / LINUX / MACOS)	ANY OS CAPABLE OF RUNNING ARDUINO IDE AND REQUIRED DRIVERS.

Tabel 3 : Software requirements

CHAPTER 4**EXPERIMENTAL RESULTS AND ANALYSIS**

The system was tested under controlled laboratory conditions to evaluate sensor responsiveness, alert accuracy, and the overall efficiency of the SMS notification mechanism. Table 1 shows the average sensor readings under three conditions: normal air, moderate smoke, and open flame.

Test Condition	MQ-2 Smoke Sensor Output (ppm)	Flame Sensor Output (Digital/Analog)	Detection Status
Normal Environment	80–120 ppm	No flame detected	Safe
Moderate Smoke	250–400 ppm	No flame detected	Alert Triggered
Direct Flame Source	120–160 ppm	Flame detected	Alert Triggered

Tabel 4: Sensor readings under different test conditions

During smoke exposure tests, the MQ-2 sensor consistently crossed the predefined threshold within 3–5 seconds, demonstrating reliable early detection. For flame detection, the IR flame sensor responded almost instantly (<1 second), indicating high sensitivity to fire signatures. The buzzer activation was immediate, while SMS notifications were delivered within 3–7 seconds depending on network quality.

The results indicate that the dual-sensor approach significantly improves the reliability of fire detection by minimizing false alarms that may occur when only one sensor is used. The flame sensor offered rapid detection for open fire, while the MQ-2 sensor effectively identified early smoke conditions, making the system suitable for both slow-burning and fast-igniting fire scenarios. The SMS alert feature consistently functioned within acceptable delay limits, ensuring the user receives timely notifications even when off-site. Overall, the system demonstrated strong performance, validating its effectiveness as a low-cost, real-time fire safety solution.

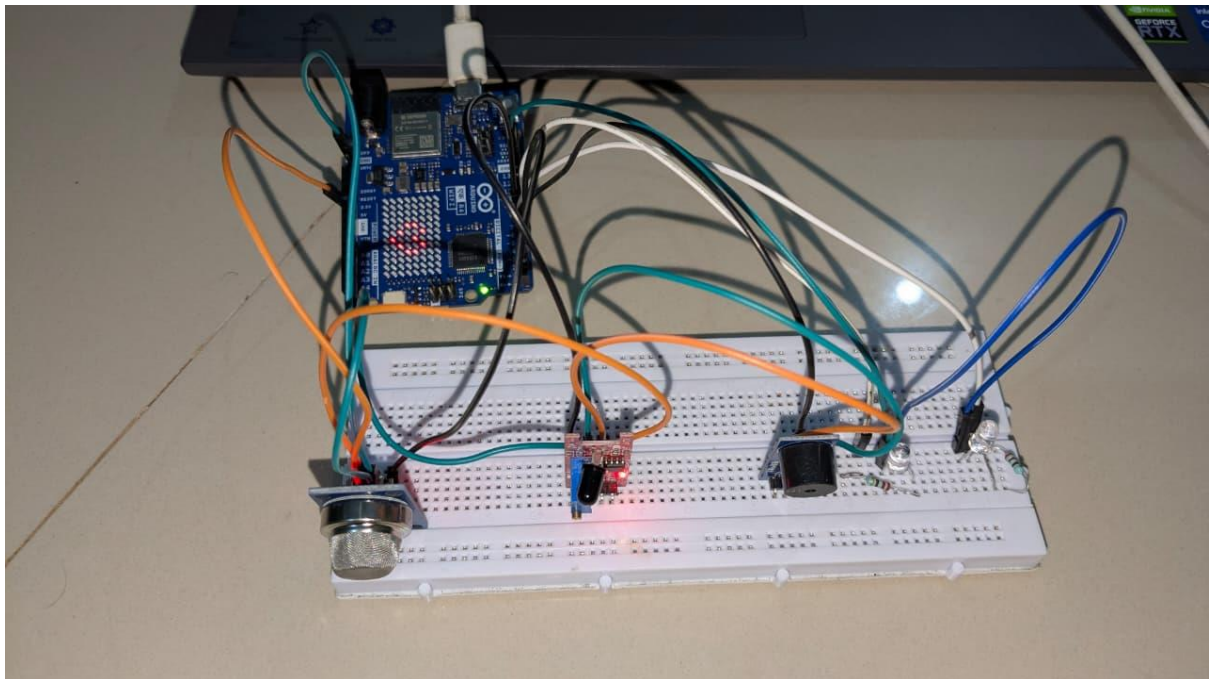


Figure 4.1: Final Implemented Prototype

The image shows the prototype of the IoT-based fire and smoke detection system built on a breadboard using an Arduino UNO R4, MQ-2 smoke sensor, flame sensor, buzzer, and indicator LEDs. All components are interconnected with jumper wires, allowing the system to detect smoke or flame and trigger both visual and audible alerts during testing.



Figure 4.2: Final SMS Alert message

This image shows the SMS alert received from the fire and smoke detection system, indicating a FIRE EMERGENCY and warning the user to evacuate immediately. It demonstrates the successful operation of the GSM module, which sends real-time notifications to the user's mobile phone during hazardous conditions.

CONCLUSION

The IoT-based Fire and Smoke Detection System developed in this project successfully provides an effective and reliable method for early detection of hazardous conditions such as smoke, gas leakage, and open flames. By integrating the MQ-2 smoke sensor, flame sensor, Arduino microcontroller, and GSM communication module, the system ensures real-time monitoring and instant alert generation. The buzzer offers immediate local warning, while the SMS notification feature allows remote users to respond quickly even when they are not physically present near the monitored area.

The results obtained under controlled laboratory conditions demonstrate that the sensors respond accurately to varying levels of smoke and flame, triggering timely alerts whenever the threshold values are exceeded. This validates the system's suitability for home safety, laboratories, warehouses, and small industrial environments.

Overall, the project highlights the importance of combining embedded systems with IoT technologies to enhance safety and prevent fire-related accidents. The system is cost-effective, easy to install, and can be expanded further with cloud monitoring, mobile applications, and automatic fire-suppression features, making it a strong foundation for advanced smart safety solutions.

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